

CyberFlow: LangFlow Cybersecurity Smart Chatbot

Implementation Documentation

Step 1. LangFlow Setup

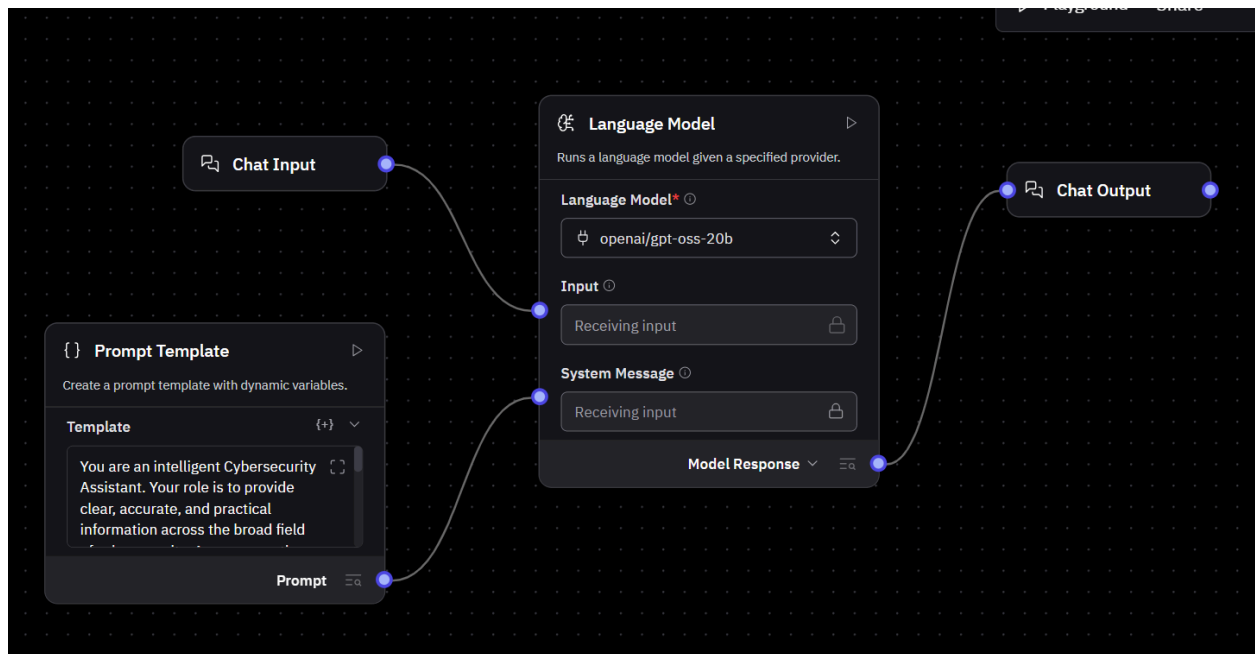
A new LangFlow flow was created for developing the cybersecurity chatbot. The following components were added:

- Chat Input
- Language Model
- Chat Output
- Prompt Template

The initial workflow was configured as:

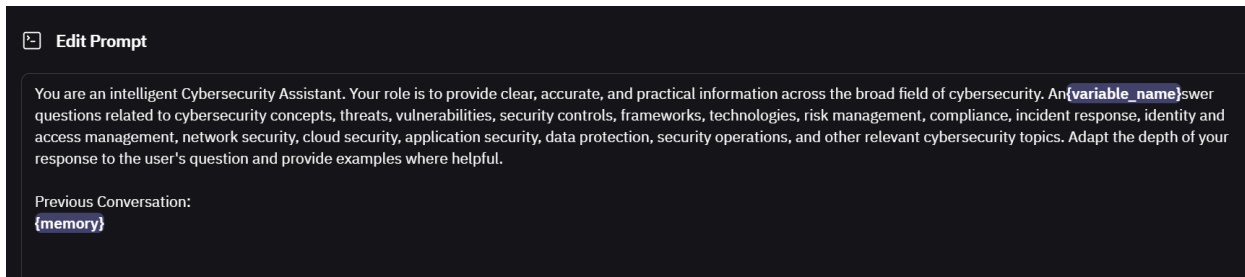
Chat Input → **Prompt Template** → **Language Model** → **Chat Output**

The Prompt Template was connected to the **System Message** input of the Language Model.



Prompt given to the Language Model.

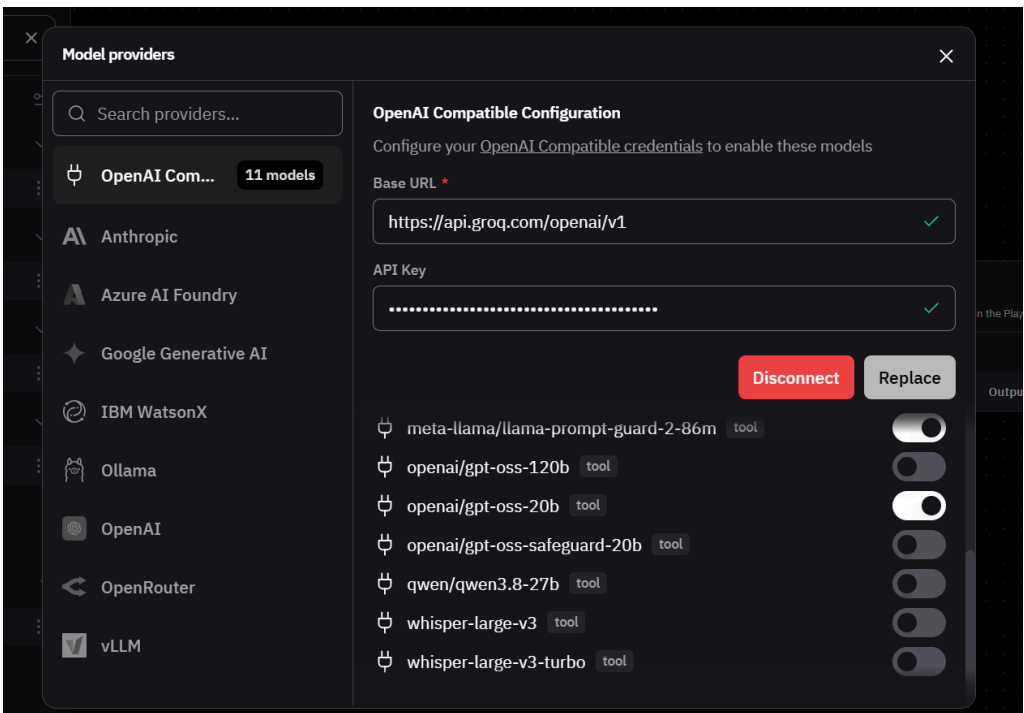
configured to define the chatbot's role as a broad cybersecurity assistant.



Step 2. Groq Model Configuration

The Language Model was configured using Groq through LangFlow's **OpenAI Compatible** provider. The following configuration was used:

Setting	Configuration
Provider	OpenAI Compatible
Base URL	https://api.groq.com/openai/v1
Model	openai/gpt-oss-20b
Tools	Disabled



Step 3. Conversation Memory

To enable context-aware conversations, **Message History** components were added to the workflow.

Message History – Retrieve

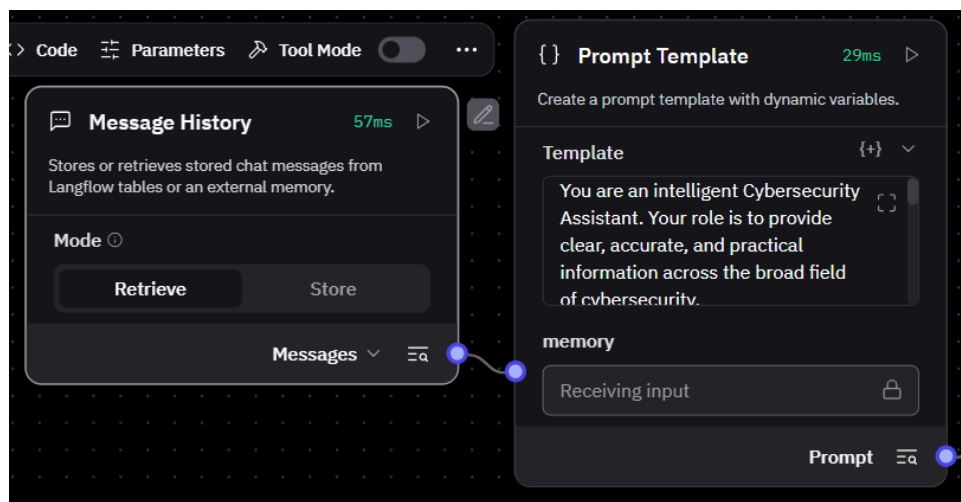
The first Message History component was configured in **Retrieve** mode. It retrieves previous messages from the current conversation session.

Prompt Template Memory

A `{memory}` variable was added to the Prompt Template:

Previous Conversation:
`{memory}`

The **Message History – Retrieve** output was connected to the `memory` input of the Prompt Template.

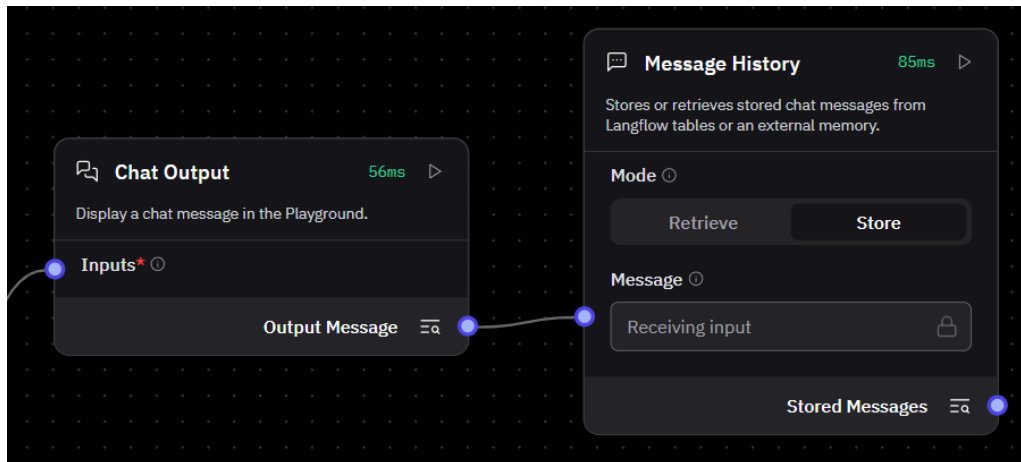


Step 4. Store Conversation History

A second **Message History** component was added in **Store** mode.

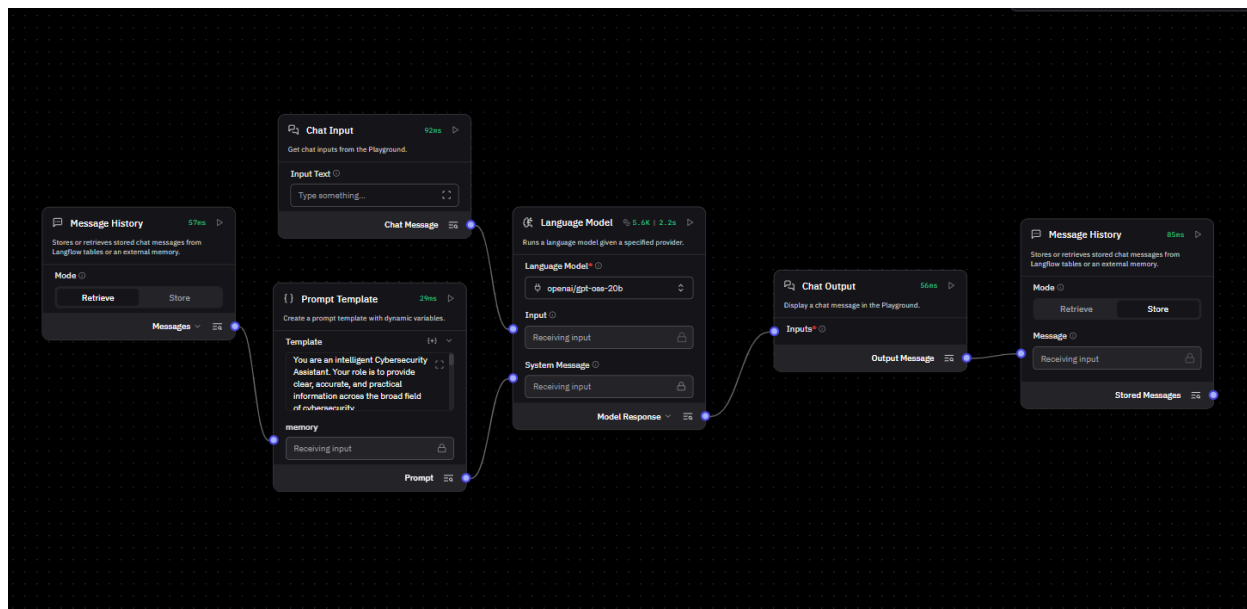
The chatbot response was connected to the Store component so that the conversation can be retained for subsequent interactions.

The Session ID was left at its default configuration for both Message History components so that the current LangFlow session can be used.



Step 5. Final Workflow

After configuring the memory functionality, the final workflow contained the following components:

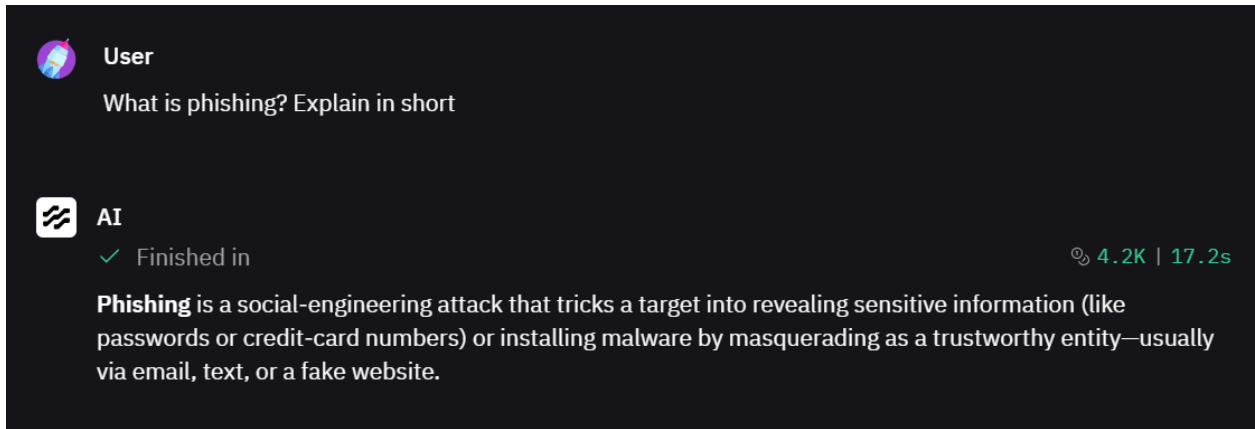


Step 6. Testing

The chatbot was tested using cybersecurity-related questions to verify both response generation and conversational context.

Test 1 – Cybersecurity Question

Query:



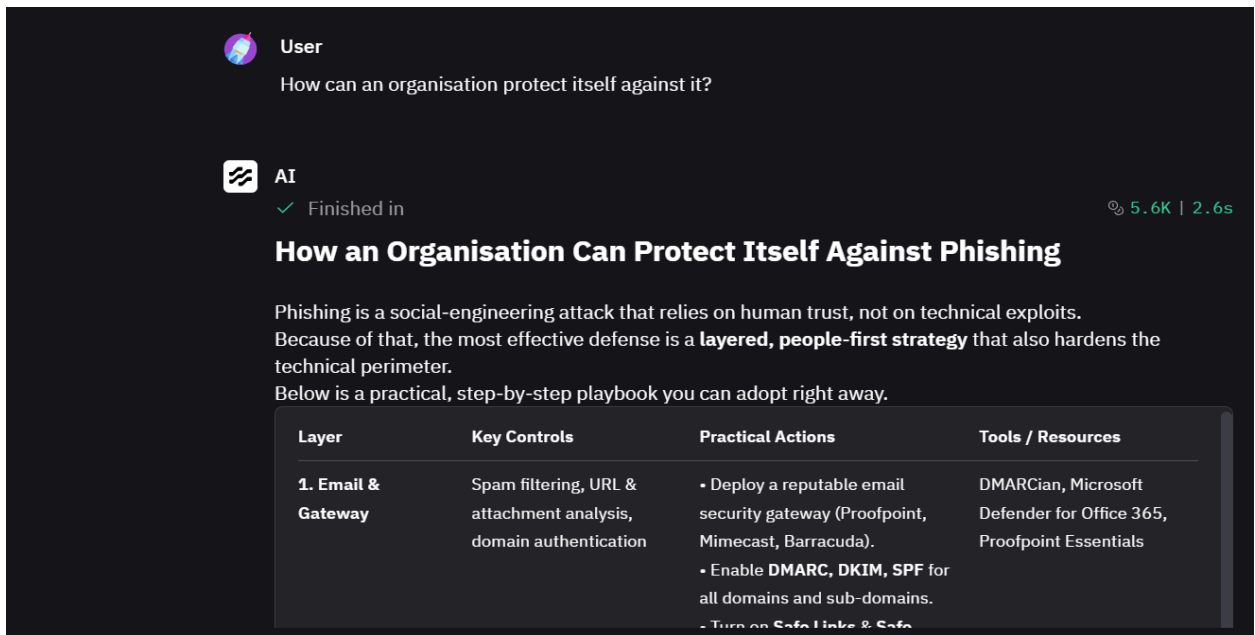
User
What is phishing? Explain in short

AI
✓ Finished in 4.2K | 17.2s

Phishing is a social-engineering attack that tricks a target into revealing sensitive information (like passwords or credit-card numbers) or installing malware by masquerading as a trustworthy entity—usually via email, text, or a fake website.

Test 2 – Follow-up Question

A follow-up question was asked without explicitly mentioning phishing again:



User
How can an organisation protect itself against it?

AI
✓ Finished in 5.6K | 2.6s

How an Organisation Can Protect Itself Against Phishing

Phishing is a social-engineering attack that relies on human trust, not on technical exploits. Because of that, the most effective defense is a **layered, people-first strategy** that also hardens the technical perimeter.

Below is a practical, step-by-step playbook you can adopt right away.

Layer	Key Controls	Practical Actions	Tools / Resources
1. Email & Gateway	Spam filtering, URL & attachment analysis, domain authentication	• Deploy a reputable email security gateway (Proofpoint, Mimecast, Barracuda). • Enable DMARC, DKIM, SPF for all domains and sub-domains. • Turn on Safe Links & Safe	DMARCian, Microsoft Defender for Office 365, Proofpoint Essentials

The chatbot correctly understood that **"it"** referred to **phishing** based on the previous conversation. This confirmed that the Message History configuration was successfully maintaining conversational context.

After successful testing, the LangFlow workflow was exported as a JSON file: **Cyberflow.json**